

United International University

B.Sc. in Computer Science and Engineering (CSE)

CSE 4889: Machine Learning

Final Exam: Fall 2025 Time: 2 Hours Marks: 40

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Answer all of the following questions **sequentially**.

1. a) Consider the following dataset representing input–output pairs: [5]

Input (X)	Output (Y)
(5,1)	7
(4,5)	9
(7,3)	10
(3,7)	8

A Generalized Regression Neural Network is to be used to predict the output corresponding to a new input value of $X = (5, 7)$. Compute the predicted output $\hat{Y}(5, 7)$ using the smoothing parameter $\sigma=0.7$.

- b) Consider the following training dataset: [5]

Input (I_1)	Input (I_2)	Desired Output (d)
1	0	0
0	1	1
1	1	1
Input (H)	Input (O)	-----
0.3	0.2	

The initial weights and biases for the hidden and output layers are given as follows:

For the hidden layer	For the output layer
$W_{OH}=0.21$	$W_{HO}=0.31$
$W_{HH}=0.11$	$W_{OO}=0.21$
$W_{1H}=-0.21$	$W_{1O}=-0.31$
$W_{2H}=0.21$	$W_{2O}=0.31$
Bias $b_H=0.51$	Bias $b_O=0.41$

Assume:

- Learning rate $\eta=0.001$
- Momentum coefficient $\alpha=0.62$

Draw a schematic diagram of the Recurrent Neural Network showing all connections, including input-to-hidden, hidden-to-output, and recurrent links in the hidden and output layers.

2. Classify the data into two classes using competitive learning. The training data [10] samples are given in the following table.

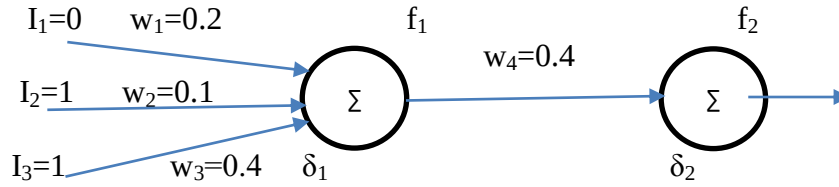
Training samples			
i1	1	1	0
i2	1	0	1
i3	0	1	0

The weight matrix is given as follows:

$$w = \begin{pmatrix} 0.5 & 0.6 & 0.2 \\ 0.6 & 0.5 & 0.8 \end{pmatrix}$$

Learning rate is 0.06. Test the unseen data sample i4= (1, 0, 0).

3. The multilayer perceptron network is given below with 4 weights: [10]



Initial weight changes are as follows: $\Delta w_1=0.2$, $\Delta w_2=-0.1$, $\Delta w_3=0.5$, $\Delta w_4=0.3$. Input training data pattern is defined as $I_1=1$, $I_2=1$, $I_3=1$, $d=1$. Determine the output Y_k for the test data pattern: $I_1=0$, $I_2=0$, $I_3=1$ where momentum co-efficient and learning rate are $\mu=0.05$ and $\eta=0.001$, respectively.

4. Consider the following sentence fragment for next-word prediction: [5+5]

“Machine learning is ”

Input weight matrix X=
$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Assume that the model consists of one self-attention layer with a vector dimension of 2, and the attention dimension is $d_k = 3$. The vocabulary for possible next token is:

{AI, a, branch, an}

Output weight matrix W=
$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

For this task, use the self-attention mechanism with parameters defined as:

$$Q=K=V=X$$

- Derive the final representation for the sequence after the self-attention layer.
- Based on the resulting context vector, predict the most probable next token from the given vocabulary.